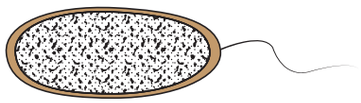
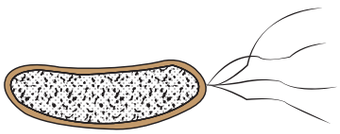
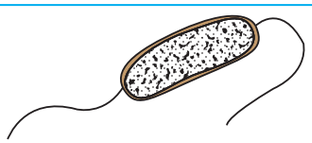
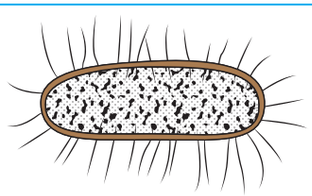
Table 7.1: Arrangement of bacterial flagella

Structure	Flagella type	Example
	Monotrichous(single flagella on one side)	*Vibrio cholera*
	Lophotrichous(tuft of flagella on one end)	*Pseudomonas fluorescens*
	Amphitrichous(single or tuft on both ends)	*Aquaspirillum serpens*
	Peritrichous(flagella throughout the cells)	*Salmonella typhi*

HOTS

A. If a bacterium loses its flagella, does it survive?

B. If you remove the cell wall from a flagellated bacterium, the organism loses the ability to move. Explain.

The presence of motility is one piece of information used to identify a pathogen in the laboratory. One way to detect motility is to stab a tiny mass of cells into soft (semi solid) medium in a test tube. Growth spreading rapidly through the entire medium is indicative of motility. Alternatively, cells can be observed microscopically by a hanging drop method.

Pili

Pili (singular pilus) are straight, short and thin and more numerous than flagella around the cell. They can be observed only by electron microscopy. They are found only in certain species of Gram negative bacteria. Pili play no role in motility. Pili originate from the plasma membrane and are made up of a special protein called pilin (Figure 7.5).

Pili play a major role in human infection by allowing pathogenic bacteria to attach to epithelial cells lining the respiratory, intestinal or genitourinary tracts. This attachment prevents the bacteria being washed away by body fluids, thus helps in establishment of infection. One specialized type of pilus (sex pilus) helps in the transfer of genetic material between the bacterial cells. This process is called conjugation.

Fimbriae

Fimbriae (singular: fimbria) is another term used for short pili that occur in great number around the cell. They enable bacteria to attach to surfaces and to each other, so that the bacteria form clumps or films called pellicles on the surface of liquid in which they are growing. Fimbriae are found in Gram positive as well as in Gram negative bacteria.

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